Anish Aggarwal

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Experience

Meta London, United Kingdom

Software Engineer *July 2024 – Present*

- Saved **$18M+/year** in infrastructure costs by designing and shipping SQL-style predicate pushdown for ML data access on Scribe, cutting network transfer **70% (4 TB/s)**; drove adoption across **10+ ML teams**.
- Designed **admission control** for Scribe readers: a **PID controller** throttling intake against CPU targets, plus **per-request memory budgeting** with live utilization tracking to reject work before OOM; eliminated resource-exhaustion incidents during traffic spikes.
- Built a pre-merge **performance-regression system** that replays sampled production requests against prod vs. candidate binaries and emits CPU/memory/network deltas, which an AI agent analyzes and posts as review comments; caught **30+ regressions** before release.

Google Warsaw, Poland

Software Engineer *May 2023 – July 2024*

- Integrated **ScaNN** approximate nearest-neighbor search into a **multi-modal retrieval pipeline**, accelerating vector search over high-dimensional embeddings under production workloads.
- Designed and shipped the **cluster lifecycle-management UI** for GKE BareMetal/VMware Clusters in Google Cloud Console, replacing CLI-only workflows with self-service provisioning and upgrades for enterprise admins.

Amazon Bengaluru, India

Software Engineer *Jan 2022 – May 2023*

- Designed a resilient backup payment flow that cut payment failures, recovering **$150K+** in the US region within 2 months of launch.
- Profiled and eliminated redundant network hops across payment services, cutting **P99 latency by 20ms** on **1M+ daily transactions**.
- Implemented asynchronous retry and fault-tolerance mechanisms, auto-recovering **34%** of transient payment failures.

LinkedIn Remote

Software Engineer Intern *May 2021 – Aug 2021*

- Cut auth-cookie signing latency **70%** on LinkedIn's member-login path by switching RSA to ECDSA; owned the project end to end through production ramp.
- Implemented ECDSA/RSA/AES-based cookie validation and renewal in **C++** and Java, and benchmarked it as a standalone executable on **Apache Traffic Server** across load profiles and crypto parameters to quantify latency regressions before rollout.

Education

National Institute of Technology Hamirpur Hamirpur, India

B.Tech in Computer Science and Engineering ***(CGPA: 9.01/10)*** *2018 – 2022*

Coursework: Distributed Systems, Database Design, Operating Systems, Networks, Algorithms.

Achievements

C++ Standards (WG21): Author of P4398R0, proposing `views::trim` / `views::trim_while` for the C++ standard library.

Elastic Contributor Program: Ranked **#1 contributor worldwide**; awarded personally by the Elastic CEO.

Open Source: Contributor to Swift-NIO (Apple), Slate, and Susi.ai (FOSSASIA).

Skills

Languages: C++ (C++17/20), Python, Java, SQL
Performance: CPU/memory profiling, multi-threading & concurrency, caching, Linux internals
Systems: Distributed systems, Kubernetes, GCP, AWS